Optimizing cardiac surgery waiting lists with machine learning: A data-driven approach for clinical prioritization

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***Abstract*—The persistence of waiting lists in universal healthcare systems poses a global challenge, driven by limited resources and increasing demand. This study presents a novel prioritization model for cardiac surgery waiting lists using machine learning (ML) algorithms trained real-world data from the Heart Institute (InCor), Clinics Hospital, University of São Paulo Medical School (HC FMUSP). Predictive models were developed using data from 588 patients who underwent valve surgery between 2010 and 2019, incorporating 44 clinical and demographic variables to predict the need for urgent surgical intervention. Multiple ML algorithms were evaluated to predict the need for urgent surgical intervention, with ensemble models showing the most promising results. The proposed model offers a data-driven, equitable, and efficient approach to surgical queue management, with potential to optimize resource allocation and improve cardiovascular outcomes.**

***Clinical Relevance*—Effective prioritization of patients on surgical waiting lists optimizes hospital resource allocation, lowers healthcare costs, and improves patient outcomes and quality of life. Integrating ML-based models into queue management offers clinical decision support by identifying high-priority patients who may benefit from earlier intervention, thereby reducing morbidity and mortality. The proposed model is designed to support, not replace, clinical judgment, serving as a data-driven tool to streamline prioritization, reduce waiting times, and decrease the incidence of urgent procedures.**

Keywords—machine learning, surgical decision making, waiting lists, prioritization component

I. Introduction

The 1988 Brazilian Constitution established healthcare as a universal right and a duty of the state, leading to the creation of the Unified Health System (Sistema Unico de Saude, SUS). This development enabled broad access to medical consultations, diagnostic exams, surgeries, and other healthcare services for population previously excluded social inequalities [1]. The expansion of SUS has brought substantial public health gains, including increased life expectancy, decreased infant and maternal mortality rates, and the near-elimination of vaccine-preventable diseases [2]. However, it has created structural challenges, such as the formation of long waiting lists for specialist care, diagnostic procedures, and elective surgeries.

The use of Machine Learning (ML) in medical diagnosis has advanced significantly [3], driven by progress in data processing and pattern recognition within clinical databases. ML offers effective tools to support physicians in disease diagnosis and clinical decision-making with improved safety and efficiency [4]. Various ML technologies have been explored for biomedical applications, particularly in diagnosis, therapeutic planning, and prognosis prediction [5].

The literature further highlights that ML techniques have, in some cases, surpassed the performance of healthcare professionals in interpreting and classifying medical images across different fields of medicine. Notable examples include the detection of diabetic retinopathy [6], classification of skin cancer [7], arrhythmia detection [8], hemorrhage identification [9], pulmonary tuberculosis classification [10], and lung nodule detection [11].

Despite these advances, the implementation of ML techniques in hospital settings remains challenging. Surgical decision-making, in particular, is influenced by a complex array of subjective factors and clinical biases that must be carefully considered [12]. Therefore, developing AI-based prioritization tools that are transparent, generalizable, and capable of reducing subjectivity in real-world hospital environments has remained a significant challenge in recent years.

Although the application of ML to surgical waitlist management remains limited in literature, recent studies have explored its potential for enhancing prioritization and resource allocation. As an illustration, [13] discusses the opportunities and limitations of AI-based approaches in improving operational efficiency and risk-based triage in public health systems, while [14] highlights how institutional factors can influence transplant waitlist outcomes and suggests that ML models may help uncover these inequities. These studies reinforce the relevance of further research on data-driven prioritization strategies in complex healthcare environments.

This study proposes a data-driven, ML-based model for prioritizing patients on cardiac surgery waiting lists. The goal is to identify patients at high-risk of deterioration to enable earlier intervention, reducing the incidence of emergency procedures. By supporting clinical judgment, the model aims to improve resource allocation, minimize delays, and enhance patient outcomes—ultimately reducing the burden of morbidity and mortality associated with prolonged surgical waiting times.

The novelty of this study lies not in algorithmic development, but in the clinical framing and application of machine learning to a rarely addressed, high-stakes prioritization problem. The focus is on curating a representative and clinically relevant dataset, defining a practical urgency label for cardiac surgery, and assessing the feasibility of integrating ML models into the real-world management of surgical waitlists within a universal public health system.

II. Materials and Methods

A. Dataset Description

This retrospective study was based on a single database provided by the Heart Institute (InCor) of the Clinics Hospital, University of São Paulo Medical School (HC FMUSP), a nationally recognized center of excellence in cardiovascular care. The dataset was fully de-identified, involved no direct patient contact, and was approved by the Institutional Review Board (IRB) under registration CAAE 39973920.1.0000.0068.

Each patient was assigned a unique primary key, enabling the integration of clinical exams records with data from emergency room visits and hospitalization logs. A structured data merging protocol ensured that each entry remained unique [15] and readily accessible for analysis. The dataset comprises records of patients who underwent cardiac valve surgery, either electively or on an urgently/emergency basis, between January 2010 to December 2019. It includes five core components: (1) data collected during emergency room admission; (2) hospitalization records; (3) information recorded at the time of cardiac valve surgery; (4) clinical examination results; and (5) prescription medication data.

All records were extracted from the InCor´s Electronic Medical Record System (SI^3) [16]. Initially, data from 5,191 patients corresponding to a total of 5,401 surgical procedures were retrieved, as some patients underwent multiple interventions during the studied period. The cohort had a mean age of 56 years (standard deviation ±14 years), with 53% of patients identified as male.

To enhance interpretability and clinical relevance, the research team held structured meetings with medical directors from the Surgical and Clinical group of Valvular Heart Disease at InCor´s Clinical Unit of Valvulopathies. These consultations guided the selection of 60 independent candidate variables for initial model training and evaluation using with ML algorithms.

The set of explanatory variables included demographic information (age, sex, body mass index), clinical history (prior cardiac surgery, comorbidities such as hypertension, diabetes, and chronic kidney disease), and functional status (NYHA class, heart failure symptoms). Laboratory parameters (hemoglobin, albumin, glycated hemoglobin, INR, creatinine) and key echocardiographic measurements (left atrial diameter, left ventricular ejection fraction, ventricular diameters, aortic and mitral valve function) were also incorporated. Medication usage (e.g., beta-blockers, anticoagulants, diuretics) and ICD-10 diagnostic codes related to valvular lesions were included as categorical inputs. Features were selected based on clinical relevance and availability, and were subject to preprocessing steps including grouping, imputation, normalization, or transformation, as previously described. A full list of clinical variables is available upon request and was reviewed in collaboration with the institution's medical team to ensure clinical relevance and consistency.

During the preprocessing phase, the original dataset of 5,191 patients was intentionally reduced to 588 cases to ensure high data quality, clear class labeling, and clinical relevance. Inclusion criteria were designed by the authors in collaboration with the hospital's clinical staff to eliminate potential biases arising from incomplete medical records, ambiguous urgency classification, and types of procedures not aligned with the study focus. Patients not under institutional follow-up prior to surgical indication were excluded, as were those underwent urgent or emergency procedures due to infective endocarditis. Class balance was achieved by selecting 294 urgent and 294 elective cases through random undersampling. A balanced dataset of 588 patients still provides sufficient statistical power, with a margin of error of 4.6% at 99% confidence and ensures that conclusions are based on clinically validated cases rather than noisy or ambiguous records. This trade-off reflects a deliberate methodological choice, prioritizing clarity and robustness over raw volume, which is aligned with best practices in feasibility studies. These and other handling procedures were conducted to prepare the dataset for model development, as detailed in the next section.

B. Preprocessing

Data preprocessing is a critical step preceding the application of ML algorithms. It involves a set of techniques designed to improve data quality, model performance and overall reliability [17].

In this study, the following preprocessing steps were performed:

- **Exclusion of specific cases:** Records of 808 patients not underwent institutional follow-up up prior surgical indication 177 patients who underwent urgent or emergency surgery due to infective endocarditis were excluded.
- **Removal of incomplete procedural codes:** A total of 2,035 surgical records lacking International Classification Disease (ICD-10) codes related to valvular anatomical lesions were removed.
- **Elimination of inconsistent data:** Thirty-five records were excluded due to inconsistencies, such as missing dates for surgical waiting list entry despite the presence of a recorded surgical procedure. After these exclusions, 2,346 valid surgical procedures remained.
- **Sample balancing:** Class imbalance is a known limitation in supervised ML, affecting performance and generalization [18]. To address this, 294 urgent surgeries were randomly selected and paired with 294 elective surgeries, resulting in a balanced dataset of 588 patients, which provides class balance and helps mitigate performance inflation seen in imbalanced datasets. The final sample offers a margin of error of 4,6% at a 99% confidence level and 3,5% at 95%. This strategy aligns with common practice in ML model training, where quality and label clarity are prioritized over raw volume, particularly in the early stages of model development.
- **Transformation of continuous variables into categorical variables:** Quantitative features such as hemoglobin, glycated hemoglobin, albumin, and body mass index (BMI) were categorized using clinically relevant cutoff points. For instance, hemoglobin was

dichotomized based on the 10 g/dL clinical threshold, which is widely used to indicate anemia severity in surgical patients. In contrast, some variables, such as the international normalized ratio (INR), were intentionally maintained as continuous to preserve potential nonlinear relationships with the outcome and to retain the granularity required for accurate clinical interpretation.

- **Normalization of numerical data:** To mitigate the effect of varying scales and magnitudes among features, numerical variables were normalized, improving training efficiency and model convergence.

- **Handling of missing data:** Missing values were addressed carefully, acknowledging that not all patients present the same clinical complaints, and that variability exists among healthcare professionals and clinical practices [18]. To improve the handling of missing laboratory and echocardiographic values, patients were grouped into five subgroups based on the anatomical classification of valvular lesions, following guidance from clinical specialists. This stratification aimed to ensure more homogeneous patient subsets, allowing for fairer imputation based on shared comorbidities and clinical patterns. This subgroup-based method preserved clinical relevance, reduced bias versus global imputation, and ensured interpretability for future use. For each group, missing numerical values were imputed using the median of patients with complete data within the same group. For example, if a patient in Group 1 lacked hemoglobin data, the missing value was imputed using the median hemoglobin value from other patients in Group 1 with available records. For medication and ICD code variables, the absence of information was interpreted as the absence of the condition or treatment requirement, rather than as missing data.

- **Dimensionality reduction:** Redundant variables were eliminated to reduce multicollinearity [18]. For instance, due to a correlation exceeding 60%, only left ventricular systolic diameter was retained among correlated diastolic diameters, and systolic volume. Similarly, due to strong association between atrial fibrillation and warfarin use, arrhythmia-related variables were excluded. In addition, some features were removed due to high missingness or inconsistent coding. For example, albumin and glycated hemoglobin (HbA1c) presented missing data rates of 23% and 29%, respectively, and were excluded from training. Several echocardiographic parameters, including valve area and systolic pulmonary artery pressure, also exhibited substantial missingness; these variables were either imputed using subgroup-specific medians or excluded when coverage was insufficient.

- **Grouping of low-frequency medication variables:** Due to the low occurrence of certain medications, drugs were grouped into pharmacological classes: captopril and enalapril were grouped as ACE inhibitors; furosemide, hydrochlorothiazide, chlorthalidone and spironolactone as diuretics; atenolol, carvedilol, and metoprolol the beta-blockers; diltiazem and verapamil as calcium channel blockers; and acetylsalicylic acid and clopidogrel as antiplatelet agents.

- **Reclassification of anatomical lesion variables**: In cases where ICD codes indicated concomitant stenosis and insufficiency were separated into two distinct variables. For example, a case coded as Rheumatic Mitral Insufficiency (I05.2) was decomposed into Mitral Stenosis (I05.0) and Rheumatic Mitral Insufficiency (I05.1).

C. *Dependent Variable*

The dependent variable in this study is the type of surgery performed, represented as a binary classification: "1" for emergency surgeries, and "0" for non-emergency (elective) surgeries. As previously described, the final dataset used for model development was balanced, with an equal distribution of classes: 50% assigned as non-emergency (0) and 50% as emergency (1).

Label validation was performed was performed by a team of clinical specialists actively involved in the project at the institution. This step was essential, as the accuracy and reliability of target variable directly impact the performance, validity, and interpretability of the ML models.

III. Machine Learning Algorithms

Given that the labels are discrete, classification algorithms were employed including Logistic Regression (LR), Decision Tree (DT), Random Forests (RF), K-Nearest Neighbors (KNN), Gradient Boosting (GB), LightGBM (LGBM) and CatBoost (CB). Additionally, a K-fold Cross-Validation was applied to strengthen the reliability of the results. The relative performance of each technique is highly context-dependent, and this work contributes by highlighting their advantages and limitations in the specific context of patient prioritization for surgical waiting lists. Other techniques may be explored in future research.

A. *Logistic Regression*

Logistic Regression (LR) is a statistical modelling technique that predicts the probability of a binary outcome based on one or more predictor variables [19]. Instead of modeling the outcome directly, LR models the log-odds of the event occurring as a linear combination of the predictors, transforming the output through a logistic function to ensure probabilities between 0 and 1. The model parameters are estimated via maximum likelihood. Classification is performed by applying a threshold (default 0.5) to the predicted probabilities, assigning labels accordingly. Logistic regression is widely used due to its interpretability, computational efficiency, and effectiveness in binary classification tasks, making it particularly suitable for binary classification problems such as the one addressed in this study.

B. *K-Nearest Neighbors*

The K-Nearest Neighbors (KNN) is non-parametric algorithm that represents each data point in a Euclidean space [20]. It classifies a sample based on the majority class among its k closest neighbors. The algorithm requires only one main hyperparameter, the number of the neighbors (k), which must be tuned to optimize classification performance. KNN is widely used in both classification and regression tasks due to its simplicity and effectiveness [20]. The optimal value of k was selected using *GridSearchCV* with 5-fold cross-validation, evaluating k in the range of 1 to 30. This procedure ensured a fair comparison with other algorithms and mitigated the risk of overfitting.

C Decision Trees

Decision Trees (DT) are hierarchical classification algorithms that partition the dataset recursively based on conditions applied to feature values. Each split is determined by a decision rule, resulting in a tree-like structure where branches represent decision outcomes, and leaf nodes represent final classes [21]. At the end of the tree, each leaf corresponds to a prediction based on the attributes considered throughout the tree path. This structure resembles human decision-making processes, making DT intuitive and interpretable.

D. Random Forest

Random Forest (RF) is an ensemble learning method that constructs multiple decision trees during training, with each tree built from a random subset of features and samples [22]. Each tree independently produces a classification, and the final prediction is made by the majority voting among all decision trees [22]. While RF are generally robust and effective, a major consideration is their tendency to overfit the training data [22], which can compromise their generalization performance on unseen data.

E. Gradient Boosting

Gradient Boosting (GB) is a robust ensemble learning technique that constructs predictive models in a sequential manner, with each new model trained to correct the residual errors of its predecessor [23]. The algorithm optimizes a predefined loss function by fitting decision trees to these residuals, thereby gradually improving predictive accuracy. Regularization techniques and tunable hyperparameters – such as the number of iterations, tree depth, and learning rate – help prevent overfitting and enhance performance [22]. Although GB models deliver strong predictive performance, they often require extensive hyperparameter tuning and can be computationally demanding, especially when applied to large datasets.

F. LightGBM

Light Gradient Boosting Machine (LightGBM) is a gradient boosting framework developed to enhance computational efficiency, scalability, and predictive accuracy, particularly in tasks involving large-scale structured datasets with high dimensionality [24]. The performance enhancements are mainly due to two significant innovations: Gradient-based One-Side Sampling (GOSS) and Exclusive Feature Bundling (EFB). GOSS reduces the computational cost of training by retaining all instances with large gradients, which have greater influence on model improvement, while randomly sampling from those with smaller gradients. This approach ensures statistical consistency and preserves generalization performance while significantly reducing training time. EFB addresses the challenge of high-dimensional sparse data by combining mutually exclusive features into a single feature without any loss of information. This reduces the number of features processed and decreases computational overhead. Together, these strategies allow LightGBM to train faster and more efficiently than traditional gradient boosting methods, without compromising predictive performance.

G. CatBoost

CatBoost (CB), short for Categorical Boosting, is a tree-based gradient boosting algorithm with the goal of overcoming two common limitations in boosting models: the inefficient handling of categorical variables and the risk of overfitting caused by target leakage, particularly in datasets with high cardinality or small sample sizes [25]. Unlike traditional methods that require manual encoding techniques such as one-hot or label encoding, CB can natively process categorical features by transforming them into statistical representations based on the target variable. To avoid information leakage, the algorithm uses an ordered permutation encoding strategy, where each instance's encoding is computed using only the preceding data points, thereby preventing the model from using its own target value in the transformation. This approach reduces overfitting and improves generalization. Additionally, CatBoost applies a smoothing term to handle infrequent categories, enhancing the stability of encodings in sparse or imbalanced datasets. These innovations allow the algorithm to achieve strong predictive performance with minimal data preprocessing, making it a robust and efficient choice for supervised learning tasks involving categorical data.

H. Classifiers performance evaluation

After applying the algorithms, their performance was evaluated and compared. The dataset was first split into training and validation (90%) and test (10%) subsets, and then within the training and validation set, it was split into 80% for training and 20% for validation. To enhance result robustness, 5-$fold$ Cross-Validation was employed.

Several standard metrics were used for model evaluation, all derived from the confusion matrix, which is a tabular representation of model performance with four entries: True Positive (TP) and False Positive (FP) in first row, and False Negative (FN) and True Negative (TN) in second row. Based on these values, classical performance metrics were calculated, including Accuracy (Acc), Sensitivity (Se), Specificity (Sp), Precision (Pr), F1-score, and Area Under the Receiver Operating Characteristic Curve (AUROC).

All analyses were conducted using Python (version 3.12.4). The implementation pipeline was built with the following open-source libraries: **Scikit-learn** (version 1.4.2) for LR, DT, RF, KNN, GB training, preprocessing utilities, and model evaluation, **Lightgbm** (version 4.6.0) for the LGBMClassifier, and **Catboost** (version 1.2.8) for the CatBoostClassifier. Additional libraries included pandas and numpy for data manipulation. The preprocessing workflow involved median imputation for missing values, normalization using **StandardScaler**, and one-hot encoding for categorical variables, all managed through a scikit-learn Pipeline to ensure consistency and reproducibility across training, validation, and testing phases. Model tuning was performed using **GridSearchCV**, and evaluation included standard metrics derived from the confusion matrix (accuracy, sensitivity, specificity, precision, and F1-score), as well as the area under the receiver operating characteristic curve (AUROC). Fixed random seeds were applied throughout the process to support reproducibility.

IV. Results

The predictive model incorporated variables across four domains: clinical comorbidities, laboratory biomarkers, echocardiographic measurements, and pharmacological treatments. Comorbid conditions included diabetes mellitus, coronary artery disease, valvular insufficiencies (aortic, mitral, and tricuspid), rheumatic aortic stenosis, and combined valvular disorders. Clinical symptoms such as dizziness and/or syncope, as well as a history of smoking, were also accounted for. Among the laboratory variables, creatinine, urea, hemoglobin, platelet count, and especially the International Normalized Ratio (INR), a key indicator of blood coagulation status and particularly relevant for cardiac patients under anticoagulation therapy, exhibited strong predictive value. Echocardiographic measurements, including left atrial diameter, left ventricular systolic diameter, valve area, ejection fraction, and systolic pulmonary artery pressure, contributed significantly to the model. Pharmacological information comprised the use of antiplatelet agents, calcium channel blockers, and the total number of prescribed medications, which served as an indirect measure of clinical complexity and disease severity. Among the evaluated models, CB and RF algorithms demonstrated the best discriminatory performance in identifying high-risk patients. On the testing subset, CB achieved a Se of 0.7517, Sp of 0,5867, Acc of 0.6678, Pr of 0.6372, F1-score of 0.6891, and AUROC of 0.6924. RF produced slightly superior results, with a Se of 0.7034, Sp of 0.6333 Acc of 0.6678, Pr of 0.6496, F1-score of 0.6754, and AUROC of 0.6997. The other algorithms tested yielded inferior performances (Table I).

A global feature importance analysis was performed for the finalist models. For the CatBoost classifier, the most relevant variables included urea, INR, age, mitral valve insufficiency, ejection fraction (Teichholz), platelet count, valve area, left atrial diameter, creatinine, pulmonary artery systolic pressure, diabetes, syncope, and coronary artery disease. In the Random Forest model, the top features were urea, INR, and platelets. These results align with clinical expectations and support the interpretability of the models.

To ensure robust evaluation of model performance and mitigate overfitting, a 5-fold cross-validation strategy was applied during training. In addition, a hold-out test set, previously unseen by the models, was used to assess generalization. This dual-validation approach enhances reliability and supports model reproducibility. Although the current study used a static dataset, future deployments could incorporate periodic model retraining using updated patient data or explore online learning techniques to allow continuous updates as new information becomes available in clinical systems.

TABLE I. Performance of the models in identifying high-risk patients ordered by F1-score (mean ± standard deviation (std)).

ML Algorithms	Subset	Accuracy (Acc)	Precision (Pr)	Sensitivity (Se)	Specificity (Sp)	F1-Score (F1)	AUROC
CB	Train	0.6654 ± 0.0086	0.6501 ± 0.0098	0.7198 ± 0.0204	0.6108 ± 0.0177	0.6830 ± 0.0105	0.7230 ± 0.0040
	Validation	0.6087 ± 0.0246	0.6016 ± 0.0249	0.6528 ± 0.0513	0.5646 ± 0.0517	0.6248 ± 0.0306	0.6416 ± 0.0263
	Test	**0.6678 ± 0.0488**	**0.6372 ± 0.0393**	**0.7517 ± 0.0634**	**0.5867 ± 0.0427**	**0.6891 ± 0.0488**	**0.6924 ± 0.0350**
RF	Train	0.6172 ± 0.0065	0.6195 ± 0.0063	0.6113 ± 0.0109	0.6231 ± 0.0049	0.6153 ± 0.0083	0.6497 ± 0.0069
	Validation	0.6125 ± 0.0259	0.6145 ± 0.0289	0.6075 ± 0.0408	0.6175 ± 0.0336	0.6106 ± 0.0325	0.6466 ± 0.0266
	Test	**0.6678 ± 0.0081**	**0.6496 ± 0.0053**	**0.7034 ± 0.0166**	**0.6333 ± 0.0001**	**0.6754 ± 0.0105**	**0.6997 ± 0.0072**
GB	Train	0.6295 ± 0.0232	0.6185 ± 0.0185	0.6821 ± 0.0400	0.5767 ± 0.0373	0.6478 ± 0.0239	0.6849 ± 0.0156
	Validation	0.5652 ± 0.0196	0.5574 ± 0.0149	0.6377 ± 0.0332	0.4923 ± 0.0212	0.5946 ± 0.0227	0.5887 ± 0.0204
	Test	**0.6203 ± 0.0461**	**0.5919 ± 0.0355**	**0.7310 ± 0.0634**	**0.5133 ± 0.0427**	**0.6537 ± 0.0469**	**0.6299 ± 0.0524**
LGBM	Train	0.6635 ± 0.0079	0.6500 ± 0.0048	0.7113 ± 0.0181	0.6155 ± 0.0090	0.6792 ± 0.0106	0.7328 ± 0.0040
	Validation	0.5859 ± 0.0403	0.5803 ± 0.0366	0.6264 ± 0.0649	0.5451 ± 0.0545	0.6008 ± 0.0445	0.6347 ± 0.0213
	Test	**0.6237 ± 0.0786**	**0.6086 ± 0.0789**	**0.6552 ± 0.0828**	**0.5933 ± 0.0880**	**0.6300 ± 0.0754**	**0.6584 ± 0.0498**
LR	Train	0.6437 ± 0.0142	0.6378 ± 0.0152	0.6689 ± 0.0125	0.6183 ± 0.0206	0.6529 ± 0.0117	0.7053 ± 0.0106
	Validation	0.6086 ± 0.0490	0.6059 ± 0.0438	0.6264 ± 0.0649	0.5908 ± 0.0557	0.6148 ± 0.0537	0.6384 ± 0.0509
	Test	**0.6203 ± 0.0258**	**0.6281 ± 0.0212**	**0.5517 ± 0.0552**	**0.6867 ± 0.0160**	**0.5862 ± 0.0404**	**0.6811 ± 0.0272**
DT	Train	0.6456 ± 0.0039	0.6242 ± 0.0156	0.7472 ± 0.0600	0.5435 ± 0.0646	0.6768 ± 0.0170	0.6877 ± 0.0052
	Validation	0.5992 ± 0.0243	0.5867 ± 0.0213	0.7057 ± 0.0891	0.4927 ± 0.1007	0.6357 ± 0.0260	0.6284 ± 0.0195
	Test	**0.5288 ± 0.0434**	**0.5266 ± 0.0392**	**0.6276 ± 0.0634**	**0.4333 ± 0.1467**	**0.5669 ± 0.0108**	**0.5726 ± 0.0446**
KNN	Train	0.6687 ± 0.0107	0.7680 ± 0.0247	0.4868 ± 0.0215	0.8514 ± 0.0222	0.5951 ± 0.0162	0.7655 ± 0.0091
	Validation	0.5709 ± 0.0112	0.6152 ± 0.0290	0.3925 ± 0.0211	0.7499 ± 0.0416	0.4775 ± 0.0133	0.6058 ± 0.0320
	Test	**0.5966 ± 0.0298**	**0.6428 ± 0.0454**	**0.4000 ± 0.0524**	**0.7867 ± 0.0293**	**0.4921 ± 0.0471**	**0.6376 ± 0.0448**

Confusion matrix analyses confirmed that CB and RF produced comparable classification outcomes. To assess whether their predictive performance differed significantly, McNemar's test was applied. The resulting p-values provided no statistical evidence to reject the null hypothesis, indicating that the models performed equivalently. Therefore, we conclude that there is no statistically significant difference between the classification performance of CB and RF, with both models demonstrating comparable predictive ability.

V. Discussion

The findings of this study demonstrate the feasibility and clinical relevance of using ML algorithms to support the prioritization of cardiac surgery waiting lists. CB and RF models exhibited superior performance in identifying patients at elevated risk for urgent surgical intervention, outperforming alternative methods such as LR and DT.

The analysis of the explanatory variables revealed a substantial degree of redundancy among clinical features, even after removal of highly correlated variables. This redundancy may have contributed to the superior performance observed in boosting-based algorithms [26], which are particularly effective in such contexts. These algorithms iteratively focus on the most informative features, reducing the influence of less relevant variables and mitigating overfitting risks. Furthermore, these algorithms combine multiple weak learners sequentially, allowing the capture of complex patterns even in the presence of multicollinearity in the data.

The identification of relevant predictive variables, such as aortic valve insufficiency, patient age, and left atrial diameter, can assist physicians in clinical decision-making and resource allocation. Notably, INR emerged as a strong predictor of urgent surgery risk, highlighting the clinical relevance of coagulation status in decision-making processes for surgical prioritization.

Comparison of these findings with previous studies [26] highlight that the use of ML algorithms for waiting list prioritization is a promising approach, with the potential to enhance both the efficiency and fairness of healthcare systems. However, it is important to emphasize that ML models are intended to complement, not replace, physicians' clinical judgment, providing objective, data-driven insights to support clinical decision-making.

Despite the promising findings, several limitations should be acknowledged. First, the dataset used was limited to a single center (InCor-HCFMUSP), which may restrict the generalizability of the findings to other institutions or healthcare systems. However, the institution involved is a national referral center for cardiovascular care in Brazil, serving a highly diverse and heterogeneous patient population. This characteristic enhances the model's potential applicability beyond the local setting, and the cohort´s demographic diversity, reflecting Brazil´s highly miscegenated population, may enhance the external validity of the model across different patient populations [27]. Second, the definition of "urgent surgery" may vary across institutions and healthcare systems, potentially complicating comparisons with other studies. Finally, the retrospective nature of the data introduces inherent limitations, such as potential selection and information biases, which may affect the accuracy and external applicability of the ML models.

Although the best-performing models in this study yielded AUROC values just below 0.70, this level of performance is not uncommon in complex clinical scenarios involving overlapping features and heterogeneous presentations. Predicting urgency in cardiac surgery is particularly challenging due to the heterogeneity of clinical presentations and shared risk profiles among patients on the waitlist. As shown by [14], even in structured referral systems, triage models achieved AUROC values between 0.71 and 0.76 across specialties, highlighting the inherent difficulty of such predictive tasks. In similar real-world applications, models with comparable discrimination metrics have been shown to provide meaningful clinical support when integrated into decision-making workflows [28].

Furthermore, the proposed model was designed to support, not replace, clinical judgment, serving as a tool to identify high-risk patients and assist in prioritization under constrained healthcare resources. Importantly, the model incorporates clinically meaningful and interpretable features, such as laboratory values, comorbidities, and hemodynamic data, which are essential for safe deployment in high-stakes environments such as surgical scheduling. In this context, even models with moderate AUROC values can offer practical value, particularly when supported by transparent feature selection and expert clinical validation.

Regarding clinical implementation, the proposed model could be integrated into electronic health record (EHR) systems through dashboards that identify high-risk patients on surgical waiting lists. Real-time updates would allow dynamic reprioritization based on new clinical data, mainly in resource-constrained environments. Alerts could be triggered to support early decision-making by multidisciplinary teams. Additionally, a decision-support workflow is proposed, including threshold calibration for risk stratification and oversight mechanisms to ensure alignment with clinical priorities. Importantly, the model is intended to function as a support tool, operating within a human-in-the-loop framework to ensure that decisions remain under clinical oversight.

VI. Conclusion

This study demonstrates the feasibility of developing a ML model to support the prioritization of surgical waiting lists, with the potential to optimize resource allocation and reduce the incidence for emergency surgeries. CB and RF algorithms exhibit string performance in identifying patients at elevated risk of requiring urgent surgical intervention, highlighting the clinical relevance of predictors such as prior cardiac surgery (reoperation), aortic valve insufficiency, patient age, and left atrial diameter.

The integration of ML algorithms with clinical expertise can enhance decision-making processes, promoting a more equitable and efficient healthcare system. It is essential to emphasize that ML models are designed to support, not replace, clinical judgment by providing objective, data-driven insights to assist in medical decision-making.

In this context, combining ML-based tools with physician expertise may lead to more effective prioritization of surgical queues, allowing for early alerts, better adjustment of waiting times for elective surgeries, and a reduction in the incidence of urgent procedures. This represents a significant step toward improving the management of surgical waiting lists in public healthcare systems.

Although this is a single-center, retrospective study, it provides valuable insight into real-world challenges associated with surgical queue management in a universal public health system. The institution involved is a national referral center for cardiovascular care in Brazil, serving a highly diverse and heterogeneous patient population. This characteristic enhances the model's potential applicability beyond the local setting. Furthermore, the study is positioned as a proof-of-concept initiative, consistent with the translational frameworks described by [28] and [29], which emphasize that institutional practices and local context can significantly influence model performance, especially in high-complexity domains such as cardiac surgery waitlisting, even in large-scale, multi-site investigations. As such, the present study represents an important step toward broader validation and real-world deployment of data-based clinical prioritization tools.

Finally, this study highlights opportunities for future research, including the evaluation of additional ML algorithms and the incorporation of novel predictive features, such as genetic markers, screening results, and lifestyle-related data. Prospective studies are also needed to assess the real-world clinical impact and scalability of ML-based prioritization models in diverse healthcare settings. Furthermore, the incorporation of prospective implementation studies within electronic health records will be essential for assessing clinical impact, scalability, and long-term sustainability.

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